



CAK2HAB3 - Dasar Kecerdasan Artifisial

REASONING

First-Order Logic

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S1 Informatika – Fakultas Informatika

Semester Genap 2024/2025



First-Order Logic

- *Propositional Logic* kurang efisien
- *First-Order Logic* lebih efisien
- *First-Order Logic* = *Predicate Logic* = *Predicate Calculus*

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- *First-Order Logic* lebih efisien
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First-Order Logic

1. **Objects:** sesuatu dengan identitas individual (*people, houses, colors, ...*)
2. **Properties:** sifat yang membedakannya dari object yang lain (*red, circle, ...*)
3. **Relations:** hubungan antar object (*brother of, bigger than, part of, ...*)
4. **Functions:** relation yang mempunyai satu nilai (*father of, best friend, ...*)



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Tata Bahasa First-Order Logic

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Sentence $\rightarrow$ *AtomicSentence*

| *Sentence* *Connective* *Sentence*

| *Quantifier* *Variable*, ... *Sentence*

| $\neg$ *Sentence*

| (*Sentence*)

AtomicSentence $\rightarrow$ *Predicate*(*Term*,...) | *Term* = *Term*

Term $\rightarrow$ *Function*(*Term*,...)

| *Constant*

| *Variable*

Connective $\rightarrow$ $\Rightarrow$ | $\wedge$ | $\vee$ | $\Leftrightarrow$

Quantifier $\rightarrow$ $\forall$ | $\exists$

Constant $\rightarrow$ *A* | *X*₁ | *John* | ...

Variable $\rightarrow$ *a* | *x* | *s* | ...

Predicate $\rightarrow$ *Before* | *HasColor* | *Raining* | ...

Function $\rightarrow$ *MotherOf* | *LeftLegOf* | ...



$Sentence \rightarrow AtomicSentence$

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$| Quantifier \ Variable, \dots \ Sentence$

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$| (Sentence)$

$AtomicSentence \rightarrow Predicate(Term, \dots) \mid Term = Term$

$Term \rightarrow Function(Term, \dots)$

$| Constant$

$| Variable$

$Connective \rightarrow \Rightarrow \mid \wedge \mid \vee \mid \Leftrightarrow$

$Quantifier \rightarrow \forall \mid \exists$

$Constant \rightarrow A \mid X_1 \mid John \mid \dots$

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Atomic sentences

- Atomic sentences dibentuk dari *Predicate(Term, ...)* atau *Term = Term*.
- Misalnya:
 - *Sepatu(Budi)*
 - *Saudara(Andi,Budi)*
 - *Memberi(Andi,Budi,KueCoklat)*
 - *Saudara(Andi) = Budi*, dan sebagainya

Complex sentences

- *Complex Sentence* dibangun menggunakan *connective*
- Contoh:
Saudara(Andi,Budi) $\Rightarrow$ Memberi(Andi,Budi,Kue)



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Inferensi pada First-Order Logic

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Inferensi pada *First-Order Logic*

- ▶ *First-Order Logic* menggunakan tujuh aturan *propositional logic*
 1. Modus Ponens
 2. And-Elimination
 3. And-Introduction
 4. Or-Introduction
 5. Double-Negation-Elimination
 6. Unit Resolution
 7. Resolution
 - Ditambah tiga aturan yang berhubungan dengan *quantifier*:
 8. Universal Elimination
 9. Existential Elimination
 10. Existential Introduction
- 



Universal Quantifiers ($\forall$)

- 
- *Universal Quantifiers* menyatakan sesuatu yang bersifat umum
 - Simbol $\forall$ (huruf A terbalik) dibaca '*For All*'
 - AnakKecil(Andi)
 - Suka(Andi,Permen)
- 



Universal Quantifiers ($\forall$)

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- ▶ *Universal Quantifiers* menyatakan sesuatu yang bersifat umum
 - Simbol $\forall$ (huruf A terbalik) dibaca '*For All*'
 - $AnakKecil(Andi)$
 - $Suka(Andi, Permen)$
 - $\forall x \text{ } AnakKecil(x) \Rightarrow Suka(x, Permen)$
- 



Existential Quantifiers ($\exists$)

- ▶ *Existential Quantifiers Menyatakan sesuatu yang berlaku sebagian*
 - Simbol $\exists$ (huruf E menghadap ke kiri) dibaca 'There Exist', yang berarti ada satu atau beberapa
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- ▶ *Existential Quantifiers Menyatakan sesuatu yang berlaku sebagian*
 - Simbol $\exists$ (huruf E menghadap ke kiri) dibaca 'There Exist', yang berarti ada satu atau beberapa
 - $\exists x \text{ AnakKecil}(x) \Rightarrow \text{Suka}(x, \text{Permen})$
- 



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Generalized Modus Ponens

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Generalized Modus Ponens

- Sangat efisien
 - Gabungkan tiga aturan inferensi: *And-Introduction*, *Universal Elimination*, dan *Modus Ponens*
- 



Generalized Modus Ponens

- Sangat efisien
 - Gabungkan tiga aturan inferensi: *And-Introduction*, *Universal Elimination*, dan *Modus Ponens*
 - Dari tiga kalimat berikut ini:
 - $AnakKandung(Wati, Budi)$
 - $SaudaraKandung(Budi, Andi)$
 - $\forall x, y, z \text{ AnakKandung}(x, y) \wedge \text{SaudaraKandung}(y, z) \Rightarrow \text{Keponakan}(x, z)$dapat dilakukan inferensi dalam satu langkah untuk menghasilkan suatu kalimat baru, yaitu: $\text{Keponakan}(Wati, Andi)$
- 



Masalah: Hukum Pernikahan

- ▶ Hukum pernikahan menyatakan bahwa suatu pernikahan adalah **tidak sah** jika kedua mempelai memiliki **hubungan keponakan**.
 - Wati menikah dengan Andi.
 - Dimana Wati adalah anak kandung Budi sedangkan Andi adalah saudara kembar Budi.
 - Dengan First-Order Logic, buktikan bahwa pernikahan Andi dan Wati adalah **tidak sah**.
- 

Langkah pertama


$$\forall x,y \text{ Keponakan}(x,y) \wedge \text{Menikah}(x,y) \Rightarrow \neg \text{Sah}(\text{Menikah}(x,y)) \quad (1)$$

$$\text{Menikah}(\text{Wati}, \text{Andi}) \quad (2)$$

$$\text{AnakKandung}(\text{Wati}, \text{Budi}) \quad (3)$$

$$\text{SaudaraKembar}(\text{Budi}, \text{Andi}) \quad (4)$$

$$\forall x,y \text{ SaudaraKembar}(x,y) \Rightarrow \text{SaudaraKandung}(x,y) \quad (5)$$

$$\forall x,y,z \text{ AnakKandung}(x,y) \wedge \text{SaudaraKandung}(y,z) \Rightarrow \text{Keponakan}(x,z) \quad (6)$$




Langkah ke dua

Dari (3.5) dan *Universal Elimination*:

$$\text{SaudaraKembar}(\text{Budi}, \text{Andi}) \Rightarrow \text{SaudaraKandung}(\text{Budi}, \text{Andi}) \quad (7)$$

Dari (3.4), (3.7), dan *Modus Ponens*:

$$\text{SaudaraKandung}(\text{Budi}, \text{Andi}) \quad (8)$$

Dari (3.6) dan *Universal Elimination*:

$$\text{AnakKandung}(\text{Wati}, \text{Budi}) \wedge \text{SaudaraKandung}(\text{Budi}, \text{Andi}) \Rightarrow \text{Keponakan}(\text{Wati}, \text{Andi}) \quad (9)$$

Dari (3.3), (3.8), dan *And-Intoduction*:

$$\text{AnakKandung}(\text{Wati}, \text{Budi}) \wedge \text{SaudaraKandung}(\text{Budi}, \text{Andi}) \quad (10)$$




Langkah ke dua



Dari (3.9), (3.10), dan *Modus Ponens*:

$$\textit{Keponakan}(\textit{Wati}, \textit{Andi}) \quad (11)$$

Dari (3.1) dan *Universal Elimination*:

$$\textit{Keponakan}(\textit{Wati}, \textit{Andi}) \wedge \textit{Menikah}(\textit{Wati}, \textit{Andi}) \Rightarrow \neg \textit{Sah}(\textit{Menikah}(\textit{Wati}, \textit{Andi})) \quad (12)$$

Dari (3.11), (3.2) dan *And-Intoduction*:

$$\textit{Keponakan}(\textit{Wati}, \textit{Andi}) \wedge \textit{Menikah}(\textit{Wati}, \textit{Andi}) \quad (13)$$

Dari (3.12), (3.13), dan *Modus Ponens*:

$$\neg \textit{Sah}(\textit{Menikah}(\textit{Wati}, \textit{Andi})) \quad (14)$$



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Unification

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Unification

- $\text{UNIFY}(p,q) = \theta$ dimana $\text{SUBST}(\theta,p) = \text{SUBST}(\theta,q)$
- θ disebut *unifer* dari dua kalimat tersebut
- $\text{Suka}(\text{Andi},x) \Rightarrow \text{Beli}(\text{Andi},x)$ berarti “Andi beli apapun yang dia sukai.”

Unification

- 
- $Suka(Andi, Roti)$
 - $Suka(y, Apel)$
 - $Suka(y, Masakan(y))$
 - $Suka(x, Wortel)$

Pengaplikasian unifikasi *premise* atau *antecedent* menghasilkan:

- $UNIFY(Suka(Andi, x), Suka(Andi, Roti)) = \{x/Roti\}$
 - $UNIFY(Suka(Andi, x), Suka(y, Apel)) = \{x/Apel, y/Andi\}$
 - $UNIFY(Suka(Andi, x), Suka(y, Masakan(y))) = \{y/Andi, x/Masakan(Andi)\}$
 - $UNIFY(Suka(Andi, x), Suka(x, Wortel)) = fail$
- 

Unification

- 
- $Suka(Andi, Roti)$
 - $Suka(y, Apel)$
 - $Suka(y, Masakan(y))$
 - $Suka(x, Wortel)$

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Generalized Resolution

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Generalized Resolution

- Resolution bisa dinyatakan sebagai Disjunctions dan Implications
- Generalized Disjunctions

Untuk p_i dan q_i , di mana $\text{UNIFY}(p_j, \neg q_k) = \theta$, maka:

$$p_1 \vee \dots p_j \dots \vee p_m$$

$$q_1 \vee \dots q_k \dots \vee q_n$$

$$\frac{}{\text{SUBST}(\theta, (p_1 \vee \dots p_{j-1} \vee p_{j+1} \vee p_m \vee q_1 \vee \dots q_{k-1} \vee q_{k+1} \dots \vee q_n))}$$

Generalized Resolution

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Untuk p_i dan q_i , di mana $\text{UNIFY}(p_j, \neg q_k) = \theta$, maka:

$$\frac{p_1 \vee \dots p_j \dots \vee p_m \quad q_1 \vee \dots q_k \dots \vee q_n}{\text{SUBST}(\theta, (p_1 \vee \dots p_{j-1} \vee p_{j+1} \vee p_m \vee q_1 \vee \dots q_{k-1} \vee q_{k+1} \dots \vee q_n))}$$

- Generalized Implications

Untuk atom p_i, q_i, r_i, s_i dimana $\text{UNIFY}(p_j, q_k) = \theta$, maka:

$$\frac{p_1 \wedge \dots p_j \dots \wedge p_{n1} \Rightarrow r_1 \vee \dots \vee r_{n2} \quad s_1 \wedge \dots \wedge s_{n3} \Rightarrow q_1 \vee \dots q_k \dots \vee q_{n4}}{\text{SUBST}(\theta, (p_1 \wedge \dots p_{j-1} \wedge p_{j+1} \vee p_{n1} \wedge s_1 \wedge \dots s_{n3} \Rightarrow r_1 \vee \dots r_{n2} \vee q_1 \vee \dots q_{k-1} \vee q_{k+1} \dots \vee q_{n4}))}$$



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Representasi Pengetahuan First-Order Logic

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Dunia *Wumpus*

▶ $\forall s,b,u,c,t \text{ Percept}([s,b,Glitter,u,c],t) \Rightarrow \text{Action}(\text{Grab},t)$





Dunia *Wumpus*

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Dunia Wumpus

▶ $\forall s,b,u,c,t \text{ Percept}([s,b,Glitter,u,c],t) \Rightarrow \text{Action}(Grab,t)$

$\forall b,g,u,c,t \text{ Percept}([Stench,b,g,u,c],t) \Rightarrow Stench(t)$

$\forall s,g,u,c,t \text{ Percept}([s,Breeze,g,u,c],t) \Rightarrow Breeze(t)$

$\forall s,b,u,c,t \text{ Percept}([s,b,Glitter,u,c],t) \Rightarrow AtGold(t)$





Dunia Wumpus

▶ $\forall s,b,u,c,t \text{ Percept}([s,b,Glitter,u,c],t) \Rightarrow \text{Action}(\text{Grab},t)$

$\forall b,g,u,c,t \text{ Percept}([Stench,b,g,u,c],t) \Rightarrow Stench(t)$

$\forall s,g,u,c,t \text{ Percept}([s,Breeze,g,u,c],t) \Rightarrow Breeze(t)$

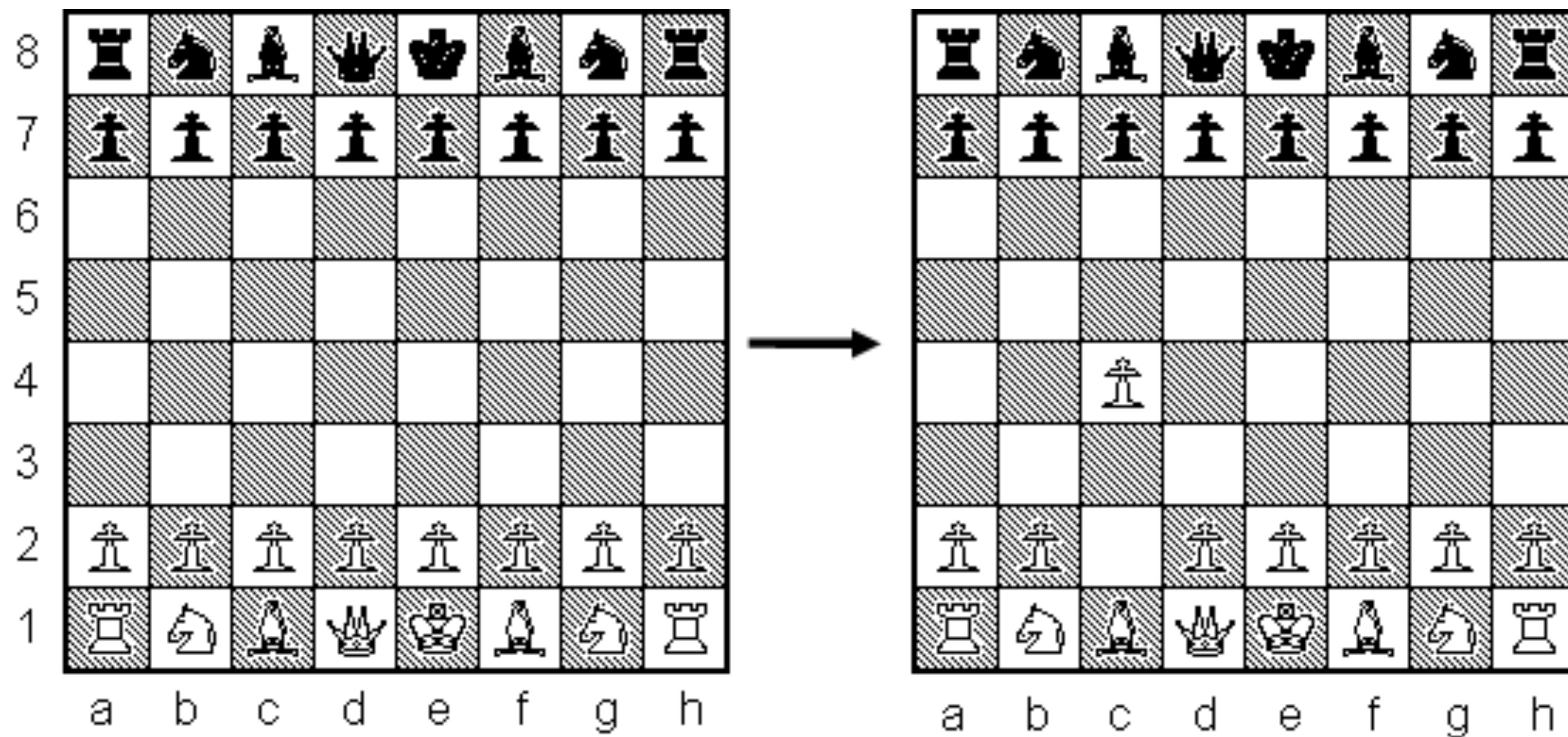
$\forall s,b,u,c,t \text{ Percept}([s,b,Glitter,u,c],t) \Rightarrow AtGold(t)$

$\forall t AtGold(t) \Rightarrow \text{Action}(\text{Grab},t)$



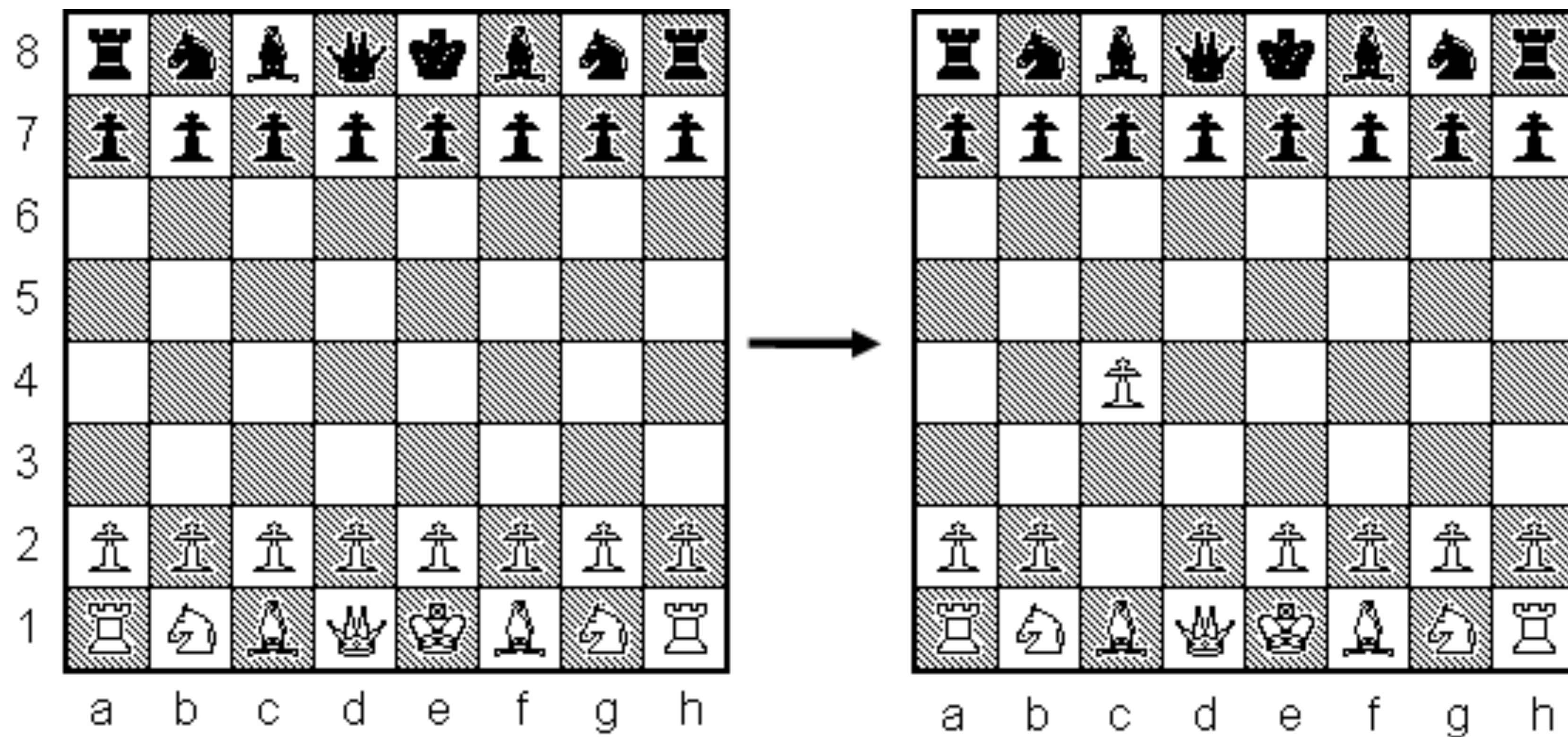


Permainan Catur





Permainan Catur





Awalnya, langkah untuk Putih

▶ 1. $a2(PP) \wedge \text{Kosong}(a3) \Rightarrow \text{Gerakkan}(PP, a2, a3)$

2. $a2(PP) \wedge \text{Kosong}(a3) \wedge \text{Kosong}(a4) \Rightarrow \text{Gerakkan}(PP, a2, a4)$

...

6. $c2(PP) \wedge \text{Kosong}(c3) \wedge \text{Kosong}(c4) \Rightarrow \text{Gerakkan}(PP, c2, c4)$

...

19. $g1(KP) \wedge \text{Kosong}(f3) \Rightarrow \text{Gerakkan}(KP, g1, f3)$

20. $g1(KP) \wedge \text{Kosong}(h3) \Rightarrow \text{Gerakkan}(KP, g1, h3)$





Terima Kasih

